

SALİH İBRAHİM USLUCAN

Automation & Software Engineer · Fountains & Show Systems · Remote

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Electronics & communication engineer (Kocaeli University, 2026) working where software meets real hardware and real customers — musical fountains and interactive water features. Sole developer of the software the company runs on daily, and of a browser engine that shows a customer their fountain before it is built. Runs LLM agents with production discipline: spec-driven generation, golden-test freezes, lint gates.

SKILLS

Programming: Python · TypeScript · C · C++ · JavaScript

AI & Automation: LLM tool-use & RAG agents · LangGraph · LLM evals · n8n workflow automation & custom nodes · OCR (Tesseract) · TDD · Git

Backend & Data: FastAPI · Flask · Docker · MCP · Supabase · Postgres/pgvector

CAD, 3D & Hardware: ezdxf · AutoCAD · DXF parsing · three.js / WebGL2 · DMX512 · RDM (E1.20)

Languages: English (fluent, C1) · Turkish (native) · Kazakh (B2)

WORK EXPERIENCE

Sales & Automation Engineer · Aquatronic — Musical Fountains

Sep 2025 – Present

- Sole developer of the company's internal software — an AI-orchestrated CAD engine, a multi-user ERP, a native C++ DMX/RDM manager, and a CRM/OCR pipeline — all in daily production use.
- Own the technical side of international sales: qualify and source new leads, prepare quotations, run the CRM, and join customer and engineering meetings in English.
- Wire and assemble electrical control panels on the production floor, and plan the montage.

Continuous Improvement & Maintenance Intern · Rokeron (formerly Formiva)

2025

- Applied 5S, Kaizen, VSM and SMED methodologies; performed preventive maintenance and fault diagnosis in industrial manufacturing workflows.

Operations Manager · Supervisor · Sales Consultant · Mr. Panda · Zuma · FLO

2021 – 2024

- Launched a restaurant's first online ordering channels (Trendyol, Yemeksepeti, Getir) and ran daily operations; earlier, scheduling, staff reviews, floor sales and customer service in retail.

PROJECTS

Show Studio — Fountain Show Preview Engine · three.js · WebGL2 · DXF · Web Audio

2026

- Browser preview engine for musical fountain shows: import the customer's DXF site plan (streaming worker parse — 2 MB+ files without freezing the UI), place catalogue hardware, generate choreography from music (FFT beat grid, section detection), play it in 3D and export a branded video. ~30 modules, 522 tests, no framework, no CDN. Live demo: salihibrahimuslucan.github.io/fountain-show-studio

Control-Panel CAD Automation · Python · TypeScript · LLM agents · ezdxf

2026

- AI-built Python engine — 800+ tests, 20+ golden projects, lint gate — turning a component netlist into rule-checked DWG schematics; drafting time cut to ~1/5. Open-source spin-off adds a FastAPI service, browser demo, MCP server and a TypeScript n8n community node. Live demo: spec2schematic.onrender.com

AI Microservice Stack · FastAPI · Docker · Postgres/pgvector · LangGraph

2026

- Containerized service: LLM tool-use agent loop, FastAPI REST layer, RAG over pgvector with cited sources, Docker Compose deployment — provider-agnostic (OpenAI/Gemini), mock-testable offline, with an LLM eval harness (golden cases, CI regression gate); the same agent is also ported to LangGraph and run side by side under a parity suite. Code: github.com/salihibrahimuslucan/ai-service-stack

AquaCONFIG v2 · C++ · Win32 · RDM

2025–2026

- Native Windows DMX/RDM device manager (Win32, GDI) with a full ANSI E1.20 RDM stack and custom PIDs; diagnosed a field PowerBox-repeater fault that was blocking RDM.

Aquatronic ERP · Supabase · Postgres · JS

2025

- Multi-user ERP — inventory, production orders, BOM consumption, CRM — on Supabase Postgres with row-level security; 8 people use it daily. Code: github.com/salihibrahimuslucan/aquatronic-erp

EDUCATION

Electronics & Communication Engineering · Kocaeli University

2019 – 2026

Computer Science · Politecnico di Torino

2021 – 2022

VOLUNTEERING

Board Member · KAZAKDER

2022 – Present

- Presented at the Kazakhstan Trade Center opening; co-organized the 2nd Unity & Solidarity Festival.